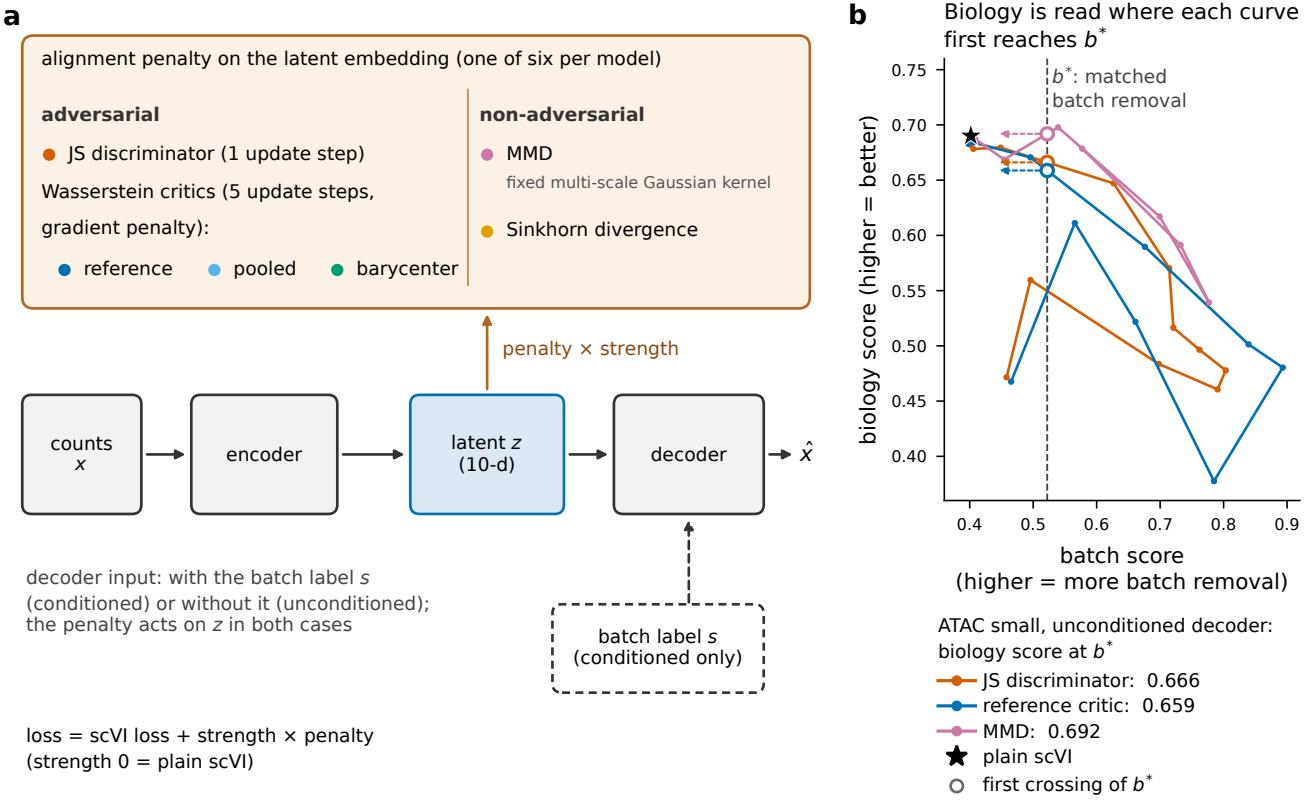

a
alignment penalty on the latent embedding (one of six per model)
adversarial
JS discriminator (1 update step)
Wasserstein critics (5 update steps, gradient penalty):
reference
pooled
barycenter
non-adversarial
MMD
fixed multi-scale Gaussian kernel
Sinkhorn divergence
penalty × strength
counts x
encoder
latent z (10-d)
decoder
x̂
decoder input: with the batch label s (conditioned) or without it (unconditioned); the penalty acts on z in both cases
batch label s (conditioned only)
loss = scVI loss + strength × penalty
(strength 0 = plain scVI)
b
Biology is read where each curve first reaches b*
b*: matched batch removal
biology score (higher = better)
batch score (higher = more batch removal)
ATAC small, unconditioned decoder:
biology score at b*
JS discriminator: 0.666
reference critic: 0.659
MMD: 0.692
plain scVI
first crossing of b*